

EXERCISE 2: MEMORY MATCHING FLUTTER GAME

COMP5450 Mobile Programming — Liang Mingchen

GitHub: <https://github.com/achenachena/comp5450-exercise-2-memory-match>

1. Overview

This solution is a Flutter/Dart memory matching game. The board is 4×5 (20 cards, 10 animal pairs). Players flip two cards per turn, earn a move when two cards are compared, and win when all pairs match.

2. Configure (first-time setup)

- Install Flutter SDK (stable) from <https://docs.flutter.dev/get-started/install>
- Install Android Studio with Flutter and Dart plugins
- Run flutter doctor and fix reported issues
- Open the exercise-2 project folder in Android Studio (File → Open)
- Run flutter pub get in the project root

3. Run the solution

- cd exercise-2
- flutter pub get
- flutter run (select Android emulator, iOS simulator, or physical device)
- Optional APK: flutter build apk --debug

4. Exact project structure

```
exercise-2/  
lib/main.dart  
lib/models/memory_card_data.dart  
lib/game/memory_game.dart  
lib/screens/memory_match_screen.dart  
lib/widgets/memory_card_tile.dart  
test/widget_test.dart  
test/memory_game_test.dart  
android/ ios/ web/ linux/ macos/ windows/  
pubspec.yaml  
README.md  
README.pdf  
screenshot-home.png  
screenshot-play.png
```

5. Features demonstrated

- Memory Match title bar with move counter badge
- Find all matching pairs! subtitle

- Colorful card backs with ? and animal emoji faces
- Match / mismatch logic with flip-back delay
- Restart button and win dialog
- flutter analyze clean; widget and unit tests pass

6. Screenshots

Home screen (face-down cards)



Gameplay (flipped cards)



Memory Match

1

Find all matching pairs! 🧠

Matched: 0 / 10



🔄 Restart